

Computer Science Revision Notes: Database Management Systems

A Database Management System DBMS is software designed to store and manage data.

Relational databases organize data into tables consisting of rows and columns.

SQL stands for Structured Query Language used to query and manipulate data.

Primary keys uniquely identify each record in a database table.

ACID properties ensure reliable processing of database transactions.

Atomicity guarantees that all operations in a transaction succeed or none do.

Consistency ensures that the database remains in a valid state.

Isolation prevents concurrent transactions from interfering with each other.

Durability guarantees that committed transactions are saved permanently.